VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 4 – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. Give two examples of **geometric** and two of **semantic** information extractable from a street image. (5 pts)

1.2. Distinguish **noise** from **outliers**. Which tools handle each? (5 pts)

1.3. State the five properties of 2-D linear transformations (Lec07). Which one does translation violate? (5 pts)

1.4. Why does SIFT compare each DoG sample with **26** neighbours instead of just 8? (5 pts)

Problem 2. Image Filtering

2.1. (a) Prove discrete convolution is commutative: $(K * I)[n] = (I * K)[n]$. (b) N images of the same static scene each carry independent zero-mean noise of variance σ^2. Prove the noise standard deviation of their average is $\sigma/\sqrt{N}$. (12 pts)

2.2. Image $I = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$. (a) Define the integral image S (with a leading zero row/column) and compute the entries needed to get the sum of rows 1–2, columns 1–2 with 4 lookups. (b) Why does an integral image make NSSD/NCC template matching fast? (13 pts)

Problem 3. Feature Detection and Matching

3.1. (a) Prove the Harris response $R = \det M - k(\operatorname{tr} M)^2$ is unchanged when the image is rotated (assume a circular Gaussian window). (b) Explain why Harris is **not** scale-invariant, with an example. (13 pts)

3.2. (a) Why is the SIFT descriptor 128-dimensional? (b) Prove the descriptor (before clamping) is invariant to $I \to aI + b$, $a > 0$. (c) What is the purpose of clamping at 0.2 and renormalizing? (12 pts)

Problem 4. Fitting and Image Transformations

4.1. (a) Derive the normal equations of least squares $\min_{\mathbf{w}} \|Y - X\mathbf{w}\|^2$. (b) For total least squares $\min \sum (ax_i + by_i - d)^2$, $a^2 + b^2 = 1$, prove the optimal line passes through the centroid and (a, b) is the eigenvector of the smallest eigenvalue of the scatter matrix. (12 pts)

4.2. (a) Prove the composition of two affine maps is affine. (b) Find the inverse of $\mathbf{p}' = A\mathbf{p} + \mathbf{b}$. (c) With $A = \operatorname{diag}(2, 4)$, $\mathbf{b} = (4, -8)$, which point maps to $(10, 0)$? (d) Why does the shear $\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$ keep every horizontal line in place? (13 pts)

Problem 5. System Design

5. An **augmented-reality** app: detect a specific book cover in phone video and overlay content exactly on the cover, even when the book is rotated, tilted, or partly covered by a hand. Design the system. (5 pts)